2023

Juniper Networks Saves Customers Millions of Dollars Using AWS Lambda

Learn how Juniper Networks reduced costs by 30 times with AI-driven solutions built on AWS Lambda and AWS IoT Greengrass.

Overview | Opportunity | Solution | Ou

30x reduction

in costs

20% targeted reduction

in mean time to resolution

Millions saved

in customer dollars

Scaled

globally

Built

a secure solution

Overview

AI-driven networking company Juniper Networks (Juniper) envisioned a transformative improvement to its customers' overall support and operational experience. Network teams often have limited insight into device states, which makes it laborious to troubleshoot and manage the entire lifecycle of network devices. "These are critical networks, so outages can cost millions per minute of downtime," says Tommy Baum, director of architecture and engineering at Juniper.

Juniper addressed this challenge by designing an innovative serverless architecture that could be implemented using Amazon Web Services (AWS) offerings. The architecture creates a concise framework for industrial Internet of Things (IIoT) designs that are hyperfocused on global telecom and mobile networks. Juniper's first use of the overall architecture resulted in a new solution, Juniper Support Insights (JSI), which elevates the customer experience by securely automating the customizable collecting and reporting of customer device data. "We saw AWS providing serverless solutions, and we thought, 'This is exactly what we were looking for,'" says Baum. Now Juniper has reduced costs by 30 times and is targeting savings of more than 5,000 percent while scaling the secure solution to the largest global deployments.

Opportunity | Using AWS Lambda and AWS IoT Greengrass to Achieve AI-Driven Networking for Juniper

Founded in 1996 in Silicon Valley, Juniper develops AI-driven networking solutions used around the world. Its engineering teams must continuously innovate while meeting stringent constraints to deliver products and services that provide compelling value to its customers, including the majority of Fortune 500 companies and the world's top 40 cloud and service providers. Because some companies use systems that date back to the early 2000s, new solutions must accommodate older tech while delivering modern functionality.

One of the challenges of running a critical, revenue-generating global network is resolving issues while minimizing downtime. Without device-level insight, engineers must assess massive networks that have tens of thousands of devices on them to pinpoint issues. "Previously, our engineers had to go back and forth with customers to identify problems, share network information, and develop solutions," says Baum.

Juniper discovered new ways to deliver dramatic operational improvements through an ambitious project for a large service provider. A team of 15 engineers worked hand in hand with the customer to implement a deeply connected solution that provided insights that drove operational efficiencies to new levels. After 2 years of intense collaboration, both the customer and Juniper were excited by the results. The challenge was to cost-effectively scale the project's capabilities across all customers while meeting a set of exceptionally complex requirements.

To unlock scale while providing comprehensive security for customers, Juniper chose AWS IoT Greengrass, an open-source edge runtime and cloud service, as a foundation for device software at the edge. And t ss architecture for its IoT solutions, Juniper pairs AWS IoT Greengrass with AWS Lambda, a serve n compute service. By using AWS services, Juniper creates a secure connection from the cloud to components of its system without leaving residual data on the edge device. And because AWS Lambda works on a pay-as-you-use model, Juniper delivers a solution in which value scales with cost. “This is virtually the first time that a company can actually achieve the objective that we’ve all been working toward as an industry for over 20 years,” says Baum. Juniper is now delivering JSI to its global customers.

Solution | Accelerating Volume While Keeping Customer Costs Flat Using AWS IoT Greengrass and AWS Lambda

Over the first 12 months following Juniper’s launch of JSI, in late 2022, the company has focused on onboarding new customers. Juniper has seen that the solution is making a big impact. Customer feedback confirms that the solution is simple to deploy and use, as well as cost-efficient to maintain, which has been a major focus area for Juniper. The company aims to reduce mean time to resolution by at least 20 percent by empowering customers, partners, and Juniper teams with more information.

Using JSI, both Juniper and its customers get detailed analytics with which to proactively predict issues, plan device end-of-life operations, and keep critical infrastructure running. “A large customer who used JSI estimated they avoided 10 outages, which would have cost millions,” says Devin Petrino, product manager at Juniper. “We are saving time when our customers need our assistance, and customers are saving time and money on their planning.” In fact, customers’ costs have been flat even as they use the advanced capabilities of JSI.

By focusing on a pervasively serverless architecture and using AWS Lambda as a key element of the implementation, Juniper can securely automate the collection and reporting of customer device states, increase productivity across multiple organizations, and reduce the total cost of ownership. The serverless architecture has changed the way that Juniper engineers go about their work. “There’s something about serverless infrastructure that’s not readily apparent to people: it creates discipline in your development team,” says Baum. “It imposes design constraints and best practices that need to be adhered to if you want to get to the scale that we operate at.” By using AWS Lambda, Juniper has followed a highly consistent set of design patterns to minimize the tedium of constant code reviews or the nuisance of oversight due to unnecessary variance.

Outcome | Meeting the Operational Needs of Complex Global Enterprises Using AWS

In 2023, Juniper focused on implementing JSI for all its customers. In 2024, the company wants to support the unique use cases of each customer and deliver self-service customization through its full suite of post-sales service offerings. And with the new data insights now at its fingertips, the potential for optimization is significant. “We’ve empowered the enterprise to customize data collection, filtering, rendering, and presentation to meet the specific needs of each customer,” says Baum. “We’re not aware of another solution that can facilitate this level of engagement.”

With JSI, Juniper’s commitment to its customers’ success has paid off. The company aims to continue its pioneering work in this area for years to come. Using AWS, Juniper engineers are free to put customer requirements first. “By migrating infrastructure management to AWS, our teams can focus on delivering value,” says Baum.

About Juniper Networks

Juniper Networks (Juniper) is a Silicon Valley–based multinational networking corporation that delivers AI-driven networking solutions and services. Founded in 1996, Juniper serves the majority of Fortune 500 enterprises.

AWS Services Used

AWS Lambda

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend service without provisioning or managing servers.

Learn more »

AWS IoT Greengrass

AWS IoT Greengrass is an open-source edge runtime and cloud service for building, deploying, and managing device software.

Learn more »

More Software & Internet Customer Stories

GERMANY

HeyJobs on AWS

HeyJobs aims to make job searching and recruitment more effective for candidates and employers.

2024

ITALY

Namirial Streamlines Contract Signing with…

Namirial enhanced its eSignAnywhere solution with generative AI using Amazon Bedrock, improving engineering productivity and the user

2024

COSTA RICA

Modernizing Applications in Minutes Using Amazon Q…

Learn how Novacomp accelerated

2024

AUSTRALIA

Dovetail Enhances Customer Insights with…

Learn how Dovetail accelerates development of new generative AI products and features using Amazon Bedrock, deploying the latest large language models while reducing security and administrative overhead.

2024

Get Started

Organizations of all sizes across all industries are transforming their businesses and delivering [...]s every day using AWS. Contact our experts and start your own AWS journey today.

Contact Sales

Learn About AWS

What Is AWS?
What Is Cloud Computing?
AWS Accessibility
AWS Inclusion, Diversity & Equity
What Is DevOps?
What Is a Container?
What Is a Data Lake?
What is Artificial Intelligence (AI)?
What is Generative AI?
What is Machine Learning (ML)?
AWS Cloud Security
What's New
Blogs
Press Releases

Resources for AWS

Getting Started
Training and Certification
AWS Solutions Library
Architecture Center
Product and Technical FAQs
Analyst Reports
AWS Partners

Developers on AWS

Developer Center
SDKs & Tools
.NET on AWS
Python on AWS
Java on AWS
PHP on AWS
JavaScript on AWS

Help

Contact Us
Get Expert Help
File a Support Ticket
AWS re:Post
Knowledge Center
AWS Support Overview
Legal
AWS Careers

Amazon is an Equal Opportunity Employer: *Minority / Women / Disability / Veteran / Gender Identity / Sexual Orientation / Age.*

Language
عربي |
Bahasa Indonesia |
Deutsch |
English |
Español |
Français |
Italiano |
Português |

2